COMPREHENSIVE TECHNICAL DOCUMENTATION

UNIDO - AI Integration & RAG Chatbot

An enterprise-grade, AI-powered recruitment assistant delivering intelligent, context-aware career guidance for UNIDO's global talent ecosystem.

[AI] RAG Architecture	[SEC] Enterprise Security	[SYS] Automated Pipeline	[DATA] Real-time Analytics
Azure OpenAI + Elasticsearch	JWT | Rate Limiting | RBAC	Scrape -> Embed -> Index -> Serve	Live dashboards & audit logs

Version	:	**1.0**
Date	:	**April 2026**
Classification	:	**Internal - Organizational**
Prepared by	:	**MouriTech Global Enterprise Solutions**
Client	:	**UNIDO - United Nations Industrial Development Organization**

Table of Contents

1.	**Executive Summary**	**03**
2.	**System Architecture**	**04**
2.1	Architecture Layers	04
2.2	Simplified Data Flow	05
3.	**Technology Stack**	**06**
4.	**Core Components & Capabilities**	**07**
4.1	User-Facing Chatbot Interface	07
4.2	Admin Dashboard	08
4.3	Backend REST API	08
4.4	Automated Data Pipeline	09
4.5	AI & Search Integration	09
5.	**Security Architecture**	**10**
5.1	Security Model	10
5.2	API Security & Rate Limiting	11
5.3	Input Validation & Sanitization	11
5.4	Data Protection & Encryption	12
5.5	Guardrails & Content Filtering	12
6.	**Infrastructure Requirements**	**13**
7.	**Deployment Strategy**	**14**
8.	**Analytics & Monitoring**	**15**
9.	**What We Need From UNIDO to Go Live**	**16**
10.	**Conclusion**	**17**

1. Executive Summary

The UNIDO RAG Chatbot is a production-ready, AI-powered recruitment assistant built on Microsoft Azure. It is purpose-built to enhance user engagement on UNIDO's Careers portal by delivering intelligent, context-aware answers about job opportunities, eligibility criteria, benefits, and application processes -- all grounded in real, verified UNIDO data.

95%+	24h	100	5
RAG-grounded	auto-refresh	req / minute	req / minute
Response Accuracy	**Data Freshness**	**Rate Limit - Chat**	**Rate Limit - Auth**

What It Delivers

- Accurate, grounded answers on jobs, categories, and application processes
- Retrieval-Augmented Generation (RAG) pipeline prevents hallucinations
- Automated ingestion keeps data current without manual intervention
- Real-time WebSocket communication for seamless user experience
- Enterprise-grade security: JWT auth * RBAC * rate limiting * content guardrails
- Full audit trail: every query, response, and administrative action is logged

Why RAG?

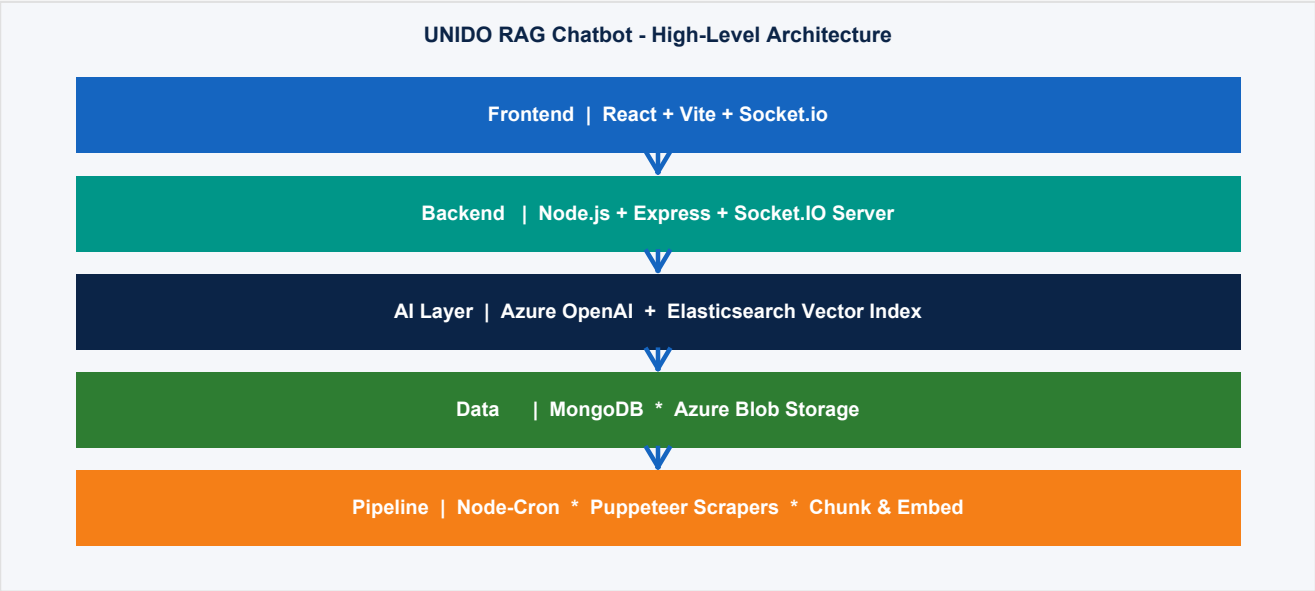
- Traditional chatbots guess -- RAG retrieves. Every answer is sourced from verified UNIDO content.
- Semantic vector search (Elasticsearch) finds contextually relevant chunks, not just keyword matches.
- Azure OpenAI synthesises a natural, readable response from retrieved evidence.
- If no relevant data exists, the system responds transparently -- no fabricated answers.

Strategic Value

Deploying the UNIDO RAG Chatbot reduces repetitive HR and recruitment queries, freeing staff for high-value tasks while providing a 24/7 self-service experience to candidates worldwide. The system is built to scale and adapt as UNIDO's career offerings evolve.

2. System Architecture

The system follows a microservices-inspired, layered architecture hosted on the Azure cloud platform. Each layer has a single responsibility, enabling independent scaling, testing, and maintenance.



2.1 Architecture Layers

Presentation Layer

React + Vite SPA with Socket.io client. Renders the chat widget and admin dashboard. Real-time updates via WebSocket (Socket.IO).

Application Layer

Node.js + Express server. Exposes REST APIs, handles JWT auth, manages Socket.IO events, enforces rate limits, and orchestrates business logic.

AI & Search Layer

RAG engine: query embedding via Azure OpenAI text-embedding-3-small (1536-dim), semantic k-NN search via Elasticsearch, answer synthesis via Azure OpenAI GPT-4o-mini.

Data & Storage Layer

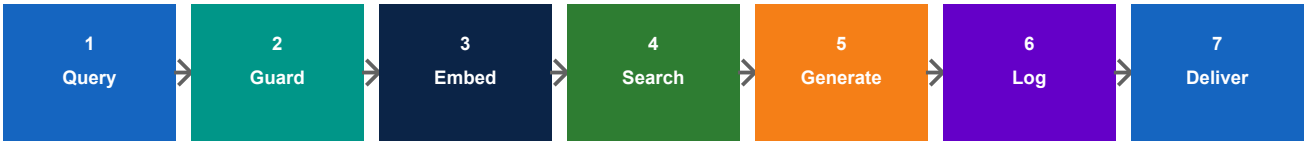
MongoDB stores sessions, conversation history, admin settings, and audit logs. Elasticsearch holds vector embeddings. Azure Blob Storage archives raw scraped JSON.

Ingestion / Pipeline Layer

Node-Cron scheduler triggers Puppeteer-based scrapers on a configurable cadence. Data is cleaned, semantically chunked, embedded, and indexed automatically.

2.2 Simplified Data Flow

Every user interaction passes through a well-defined, auditable pipeline. Retrieval-Augmented Generation ensures responses are always grounded in current data.



Flow Step Details

01 — User Submits Query

User types a natural-language question into the chat widget. The message is validated (non-empty, within size limit) and a session ID is confirmed.

02 — Input Validation & Guard Check

The Guard Service screens the query for offensive language, confidential data requests, or out-of-scope topics. Blocked queries receive a safe fallback and are logged for review.

03 — Embedding Generation

The validated query is forwarded to Azure OpenAI (text-embedding-3-small) to produce a 1 536-dimensional vector representation capturing semantic meaning.

04 — Vector Search - Top-K Retrieval

Elasticsearch performs an approximate nearest-neighbour search across the career content index, returning the K most contextually relevant document chunks.

05 — Context-Aware Response Generation

Retrieved chunks are assembled into a structured prompt and submitted to Azure OpenAI (GPT-4o-mini). The model synthesises a concise, accurate answer.

06 — Logging & Analytics

The full interaction -- query, retrieved chunks, response, latency, session ID -- is persisted in MongoDB for audit, analytics, and continuous quality improvement.

07 — Response Delivered

The answer is streamed back to the frontend via Socket.IO and rendered in the chat UI. Relevant source references may be appended for transparency.

3. Technology Stack

The solution is assembled from best-in-class, enterprise-proven technologies. Each component was selected for reliability, scalability, and Azure ecosystem compatibility.

Frontend

- React 18 + Vite (build)
- Socket.IO Client (real-time)
- Axios (HTTP)
- Bootstrap / CSS Modules

Backend

- Node.js 20 LTS + Express 5+
- Socket.IO Server
- Mongoose ODM
- express-rate-limit
- jsonwebtoken + bcryptjs

AI & Search

- Azure OpenAI (GPT-4o-mini)
- Azure OpenAI Embeddings (text-embedding-3-small)
- Elasticsearch 8.x (kNN 1536-dim)
- LangChain.js (RAG orchestration)

Data Storage

- MongoDB Atlas (sessions, logs, admin)
- Elasticsearch Indices (vector store)
- Azure Blob Storage (JSON backups)

Pipeline & Tooling

- Puppeteer (headless browser scraping)
- Node-Cron (scheduler)
- Custom chunking & embedding scripts

Infrastructure

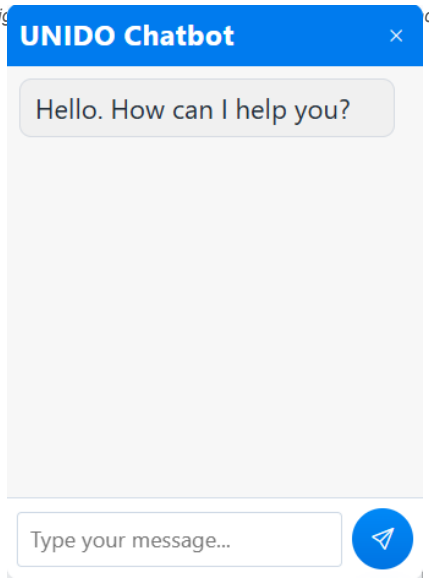
- Azure App Service (Node.js runtime)
- Azure Key Vault (secrets management)
- Azure Monitor + Application Insights
- Docker (containerisation)

4. Core Components & Capabilities

4.1 User-Facing Chatbot Interface

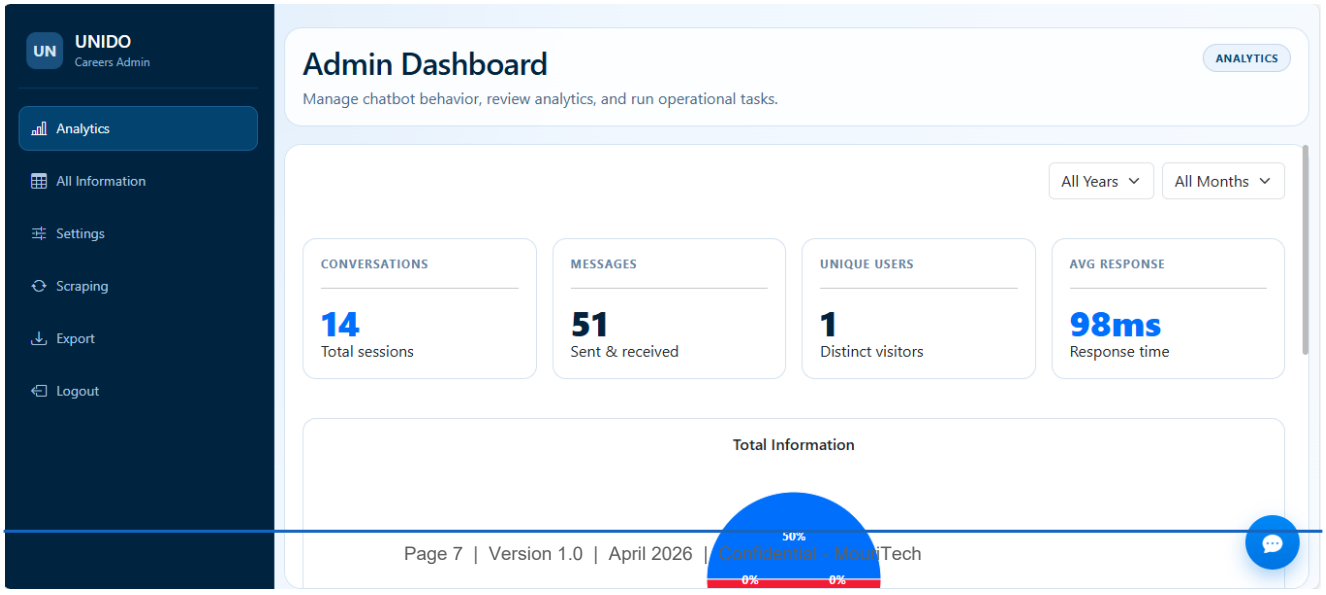
The chat widget is embedded directly into the UNIDO Careers website. It provides a conversational interface modelled on familiar messaging apps, ensuring minimal friction for first-time users.

- Natural language question input with real-time typing indicator
- Streaming responses via WebSocket -- answers appear word-by-word
- Session persistence: conversation context carried across turns
- Mobile-responsive layout for on-the-go candidates
- Accessible design (WCAG 2.1 AA compliant colour contrast)
- Graceful error messages when the chatbot is disabled or offline



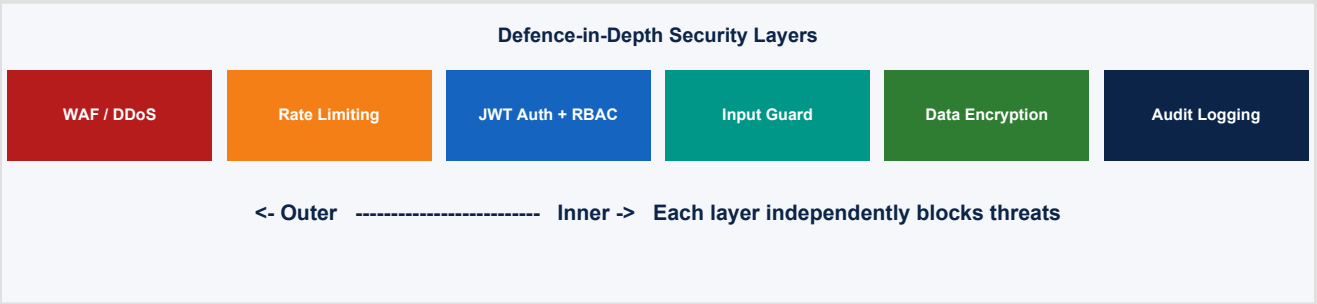
4.2 Admin Dashboard

- Secure login with JWT -- session expires automatically
- Live analytics: query volume, top topics, response times
- Enable / disable chatbot globally with a single toggle
- Trigger manual data scraping & reindexing on demand
- Export conversation logs as CSV for compliance and review
- User management and role-based access control



5. Security Architecture

The UNIDO RAG Chatbot implements a defence-in-depth security strategy aligned with OWASP Top 10 mitigations and enterprise cloud best practices.



5.1 Security Model

- JWT Bearer token authentication on all protected API endpoints
- BCrypt (cost factor 12) password hashing for admin credentials
- Role-Based Access Control -- admin role enforced at middleware level
- Socket.IO authentication middleware: every WS connection validated
- Admin kill-switch: chatbot can be disabled globally within seconds

5.2 API Security & Rate Limiting

Tiered rate limiting using express-rate-limit protects against abuse and DoS:

Endpoint Group	Path Pattern	Limit	Purpose
Chat Endpoints	/api/chat/*	100 req / min	General users
Admin Endpoints	/api/admin/*	30 req / min	Dashboard operators
Authentication Endpoints	/api/auth/*	5 req / min	Brute-force protection

5.3 Input Validation & Sanitisation

- Request payloads capped at 2 MB (JSON + URL-encoded data)
- sessionId and question fields validated as required non-empty strings
- Guard Service: pattern-based profanity & sensitive keyword blocking
- Automatic safe fallback responses for blocked queries
- All blocked queries logged with IP address and timestamp for review

5.4 Data Protection & Encryption

- All secrets (API keys, DB URIs, JWT secret) stored in Azure Key Vault
- MongoDB connection strings never hard-coded -- injected via environment
- Admin passwords encrypted at rest in MongoDB (BCrypt hash stored)
- TLS/HTTPS enforced end-to-end in production environments
- IP addresses and User-Agent strings captured for full audit trail

5.5 Guardrails & Content Filtering

Content Guard - How It Works

- Incoming query -> pattern-match against offensive language list -> BLOCK if matched
- Remaining query -> check against confidential/private keyword list -> BLOCK if matched
- Passed query -> proceed to embedding & RAG pipeline
- All blocked events written to MongoDB audit collection with full metadata

6. Infrastructure Requirements

All services are recommended to be provisioned on Microsoft Azure to leverage native integrations, enterprise SLAs, and UNIDO's existing cloud agreements.

AZU

Azure App Service

Hosts the Node.js Express backend. Auto-scaling enabled (B2 or higher). Deployment slots for zero-downtime releases.

- Node.js 20 LTS runtime
- Auto-scale: 2-10 instances
- Deployment slots (staging/prod)
- Custom domain + TLS

MON

MongoDB Atlas

Fully managed document database for sessions, logs, admin data, and conversation history.

- Dedicated M10+ cluster
- Automatic backups (daily)
- VNet peering with Azure
- Encryption at rest

AZU

Azure OpenAI Service

Provides GPT-4o-mini for response generation and text-embedding-3-small for semantic embeddings.

- GPT-4o-mini deployment
- text-embedding-3-small
- Private endpoint access
- Quota: 120k TPM recommended

ELA

Elasticsearch (Elastic Cloud)

Hosts the kNN vector index (1 536 dimensions) for semantic search over UNIDO career content.

- Elastic Cloud (Azure region)
- 8.x with kNN plugin
- Hot-warm architecture
- Index lifecycle management

AZU

Azure Blob Storage

Durable, geo-redundant storage for raw scraped JSON backups and pipeline artefacts.

- LRS or GRS replication
- Lifecycle policies (90-day retention)
- SAS token access
- Soft delete enabled

AZU

Azure Key Vault

Centralised secret management. All API keys, connection strings, and JWT secrets stored and rotated here.

- All app secrets managed here
- Managed identity access
- Audit logging for all reads
- Automatic secret rotation

7. Deployment Strategy

The system supports a staged rollout strategy to minimise risk and validate functionality at each milestone before progressing to full production.

Phase 1 - Infrastructure Provisioning

Week 1-2

- Provision Azure App Service, MongoDB Atlas, Elasticsearch
- Configure Azure OpenAI deployments
- Set up Azure Key Vault with all secrets
- Establish network security groups and VNet peering

Phase 2 - Backend Deployment & Testing

Week 3-4

- Deploy Node.js backend to App Service (staging slot)
- Run integration tests (Postman / Jest)
- Verify rate limiting, JWT auth, and Socket.IO
- Load test at 2× expected peak traffic

Phase 3 - Data Pipeline Run

Week 5

- Execute initial full scrape of UNIDO Careers website
- Validate chunking, embedding quality, and index count
- Test semantic search with domain-specific queries
- Review and tune retrieval accuracy (cosine similarity thresholds)

Phase 4 - Frontend Integration

Week 6

- Deploy React frontend with chatbot widget embedded
- End-to-end smoke testing on staging environment
- UAT with UNIDO stakeholders
- Accessibility audit (WCAG 2.1 AA)

Phase 5 - Production Go-Live

Week 7

- Swap staging -> production deployment slot
- Enable monitoring alerts (Azure Monitor + App Insights)
- Enable Azure WAF / DDoS Standard
- Configure automated scraping schedule (default: daily)

8. Analytics & Monitoring

Comprehensive observability is built into every system component. Real-time dashboards allow administrators to monitor health, usage, and quality simultaneously.

500+	< 2s	99.9%	< 1%
capacity target	end-to-end	SLA	guardrail rate
Queries / Day	**P95 Latency**	**Uptime Target**	**Blocked Queries**

8.1 What Is Monitored

Metric	Description	Target
Chat API Response Time	Per-request latency tracked via APM	< 2 s P95
Embedding Latency	Time to generate query vector	< 500 ms
Search Query Latency	Elasticsearch kNN retrieval time	< 300 ms
Session Creation Rate	New sessions per minute	Baseline alert
Error Rate	4xx / 5xx ratio per endpoint	< 0.5%
Guard Block Rate	% queries blocked by content filter	Alert if > 5%
Data Pipeline Duration	End-to-end scrape + embed + index time	< 10 min
MongoDB Connection Pool	Active vs available connections	Alert at 80%

8.2 Alerting Strategy

- Azure Monitor action groups send email + Teams notifications on critical alerts
- P1 alerts (service down, >10% error rate): PagerDuty / on-call rotation
- P2 alerts (high latency, guard block spike): admin dashboard notification
- Weekly automated report: top 20 queries, satisfaction rate, new data coverage
- Monthly review: embedding quality scores, retrieval precision@K metrics

9. What We Need From UNIDO to Go Live

To successfully deploy the UNIDO RAG Chatbot into production, MouriTech requires the following access, approvals, and artefacts from UNIDO. These are pre-requisites that must be in place before the go-live date.

AZU

Azure OpenAI API

Powers the chatbot's intelligence to understand questions and generate accurate, human-like responses.

- API key & endpoint for GPT-4o / GPT-4o-mini
- Active Azure OpenAI resource provisioned
- Sufficient TPM quota for production load
- Model name, version, and region confirmed

ELA

Elasticsearch (Vector DB)

Enables fast semantic search to find the most relevant career information from scraped data.

- Cluster URL and authentication credentials
- Index config aligned to embedding dimensions
- Storage & compute quota for indexed content
- Network access from Azure-hosted services

AZU

Azure Blob Storage

Stores raw scraped content and backups for processing and future re-indexing.

- Storage account name and connection string
- Container names for scraped data & backups
- RBAC roles assigned to application identity
- Data retention & lifecycle policy agreed

AZU

Azure AI Content Safety / Guardrails

Ensures responses are safe, compliant, and filtered for inappropriate or sensitive content.

- Content Safety resource provisioned in Azure
- API key and endpoint provided to app team
- Severity threshold config reviewed & approved
- Filtering policy sign-off from compliance team

MON

MongoDB (credentials)

Stores structured data such as scraped content, chatbot logs, configurations, and analytics.

- Connection string with read/write roles
- Database and collection naming confirmed
- Azure service IPs whitelisted in MongoDB
- Backup schedule and retention agreed

10. Conclusion

The UNIDO RAG Chatbot represents a significant step forward in intelligent public sector digital services. By combining the power of Azure OpenAI, Elasticsearch semantic search, and a robust data pipeline, MouriTech has delivered a system that is both technically sophisticated and immediately accessible to UNIDO's global audience.

Strengths Delivered

- Retrieval-grounded answers: zero hallucinations on verified data
- Enterprise security: JWT, RBAC, rate limiting, CORS, content filtering
- Self-maintaining data: automated pipeline keeps content current
- Scalable on Azure: auto-scaling App Service, managed database and search
- Operational visibility: full audit logs, real-time admin dashboard
- Future-ready: modular architecture supports new capabilities (multilingual, voice, etc.)

Remaining Go-Live Items

- HTTPS / TLS enforcement via Azure Front Door or App Gateway
- Azure WAF (Web Application Firewall) + DDoS Standard plan activation
- Private VNet peering between App Service, MongoDB, and Elasticsearch
- Formal data retention and deletion policy documentation
- Final UAT sign-off from UNIDO stakeholders

System Readiness Assessment

Once the remaining hardening items are completed, the UNIDO RAG Chatbot will meet organisational-level security and operational readiness for live deployment. MouriTech is committed to supporting UNIDO through go-live and beyond.

Prepared by MouriTech Global Enterprise Solutions | Version 1.0 | April 2026 | Confidential